

Ahmad Alamrousy

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Computational scientist and Machine Learning Engineer with 9+ years of experience in mathematical modeling, statistical inference, optimization, and large-scale machine learning systems. My work has focused on developing data-driven computational frameworks for extracting meaningful patterns from complex and noisy datasets, combining principles from applied mathematics, algorithms, and scientific computing. I am pursuing an M.S. in Data Science while strengthening my background in physics, quantum mechanics, and quantum information science in preparation for doctoral research in Quantum Science and Engineering. My long-term goal is to bridge machine learning and physics through computational approaches for modeling complex quantum and physical systems.

EXPERIENCE

Senior Machine Learning Engineer — infrashield, IL

Jan 2025 – Present

- Designed and deployed mathematical learning systems based on statistical inference, optimization, and high-dimensional representation learning, achieving **92% predictive accuracy** while maintaining robustness under changing data distributions.
- Developed physics-inspired monitoring and calibration frameworks to characterize **distribution shifts, uncertainty propagation, and model stability**, analogous to parameter drift and noise correction in continuously operating experimental systems.
- Integrated large-scale neural architectures and language models into scientific-style information extraction pipelines, applying principles from **dimensionality reduction, probabilistic modeling, and information theory** to improve automated decision processes.
- Built closed-loop machine learning systems combining automated measurement, feedback, and adaptive retraining mechanisms, enabling models to self-correct under non-stationary environments.
- Developed scalable computational pipelines using optimization techniques and numerical methods to reduce model deployment cycles from weeks to hours.

Machine Learning Engineer — Master Works, Riyadh, KSA

Jan 2022 – Dec 2024

- Developed large-scale computational models for analyzing complex educational systems using **statistical learning theory, time-series analysis, and predictive modeling** across more than **10TB of daily data**.
- Applied mathematical forecasting techniques, including temporal modeling, regression analysis, and probabilistic estimation, improving prediction accuracy of student performance by **40%**.
- Designed algorithms for extracting meaningful signals from noisy, high-dimensional datasets, applying concepts from **signal processing, statistical mechanics-inspired data analysis, and pattern recognition**.

- Engineered distributed computational frameworks for large-scale data processing, optimizing numerical workflows and reducing computational latency by **60%**.
- Built real-time analytical systems translating complex measurements into interpretable models through statistical inference and uncertainty-aware decision frameworks.

Data Scientist & Machine Learning Engineer — Unilever, Dubai, UAE Jun 2020 – Dec 2021

- Developed mathematical modeling frameworks for complex behavioral systems using statistical inference, machine learning optimization, and predictive analytics, achieving **85% classification accuracy**.
- Applied computational methods to extract latent structures from large-scale, unstructured datasets using representation learning and automated feature discovery.
- Designed machine learning algorithms for image and text analysis, leveraging concepts from **computer vision, information theory, and probabilistic modeling**.
- Implemented automated data processing pipelines transforming heterogeneous observations into quantitative models for decision-making.
- Applied rigorous model evaluation techniques, including uncertainty analysis, validation strategies, and generalization testing to improve reliability on unseen data.

EDUCATION

M.S., Data Science — Eastern University, Philadelphia, USA Jan 2024 – Apr 2026

- *Advanced study in **mathematical modeling, statistical inference, computational methods, optimization, and machine learning theory** for analyzing complex, high-dimensional systems.*
- *Applied concepts from **linear algebra, probability theory, numerical optimization, and statistical mechanics-inspired computational approaches** to develop data-driven models.*
- *Coursework and research preparation strengthened foundations in **applied mathematics, computational science, and algorithmic modeling**, providing a bridge between machine learning and physics-based simulation approaches.*
- *Currently expanding this background toward **quantum information science, quantum algorithms, and computational physics** through independent study of quantum mechanics and quantum computing frameworks.*

B.S., Computer Science — coursework concentration in Physics and Mathematics — menofiya University, menofiya, Egypt Jun 2015 – Jun 2019

- Completed rigorous coursework combining **computer science, applied mathematics, and physics**, developing a foundation in computational approaches for modeling physical systems.

- Studied core mathematical and physical concepts including **calculus, linear algebra, differential equations, electromagnetism, modern physics, and introductory quantum mechanics.**
- Applied computational thinking and mathematical methods to solve complex problems involving **numerical analysis, algorithm development, and quantitative modeling.**
- Built a foundation in the intersection of **physics, mathematics, and computation**, later extended through graduate-level work in data science, machine learning, and statistical modeling.
- Currently strengthening expertise in **quantum mechanics, quantum information theory, and quantum computing** through self-directed study and platforms such as Qiskit and advanced quantum computing courses.

TECHNICAL SKILLS

Physics, Mathematics & Computational Science Foundations

- Mathematical foundations: **Linear Algebra (matrix decomposition, eigenvalue problems), Multivariable Calculus, Differential Equations, Optimization Methods, Probability Theory, Bayesian Inference, Statistical Modeling**
- Physics foundations: **Classical Mechanics, Electromagnetism, Modern Physics, Computational Modeling of Physical Systems**
- Computational methods: **Numerical Analysis, Scientific Computing, Fourier Analysis, Signal Processing, Dynamical Systems, Time-Series Modeling, Anomaly Detection, Uncertainty Quantification**
- Mathematical approaches for complex systems: **High-dimensional data analysis, stochastic processes, parameter estimation, calibration, and model validation**

Machine Learning & Scientific AI

- Deep learning frameworks: **PyTorch, TensorFlow, Keras, scikit-learn, XGBoost**
- Machine learning methods: **Supervised/Unsupervised Learning, Representation Learning, Neural Networks, Optimization-based Learning, Probabilistic Models**
- Deep learning architectures: **CNNs, RNNs, Transformers, Large Language Models (LLMs), Fine-tuning and Retrieval-Augmented Generation (RAG)**
- Scientific machine learning concepts: **Physics-informed modeling, surrogate modeling, data-driven simulation, inverse problems, and uncertainty-aware prediction**

Data Science & Large-Scale Computational Systems

- Data processing: **PySpark, distributed computing, ETL pipeline design, streaming data architectures**
- Databases: **PostgreSQL, MongoDB, SQL-based analytics systems**
- Large-scale data analysis: **Feature engineering, statistical analysis, data pipelines, high-dimensional dataset processing**

MLOps, Scientific Computing Infrastructure & Deployment

- Cloud platforms: **AWS SageMaker, Google Cloud Platform, Azure Machine Learning**
- Infrastructure: **Docker, Kubernetes, Git/GitHub, CI/CD automation**
- Production ML systems: **Model monitoring, automated retraining pipelines, performance calibration, deployment optimization**

Scientific Programming & Data Visualization

- Programming languages: **Python, SQL**
- Scientific computing tools: **NumPy, SciPy, Pandas, Jupyter**
- Development frameworks: **Flask, Django, Streamlit**
- Visualization and analysis: **Matplotlib, Seaborn, Tableau, Power BI, SPSS, SAS, Weka**

SELECTED TECHNICAL PROJECTS

- ETEC — statistical software platform for advanced data management and business intelligence
- Time-series forecasting system for judicial caseload projection, Ministry of Justice
- Sentiment analysis pipeline for large-scale social media data
- Stock price prediction using time-series forecasting
- Blindera — assistive software enabling social media interaction for visually impaired users

CERTIFICATIONS & ACHIEVEMENTS

- Certified Scrum Master (CSM)
- 5th Place, Google-affiliated ICPC Problem-Solving Competition
- Best Employer of the Year, Master Works — for statistical software development
- Market Analysis Certification, Virginia University; Data Science Specialization, Wesleyan University; Advanced Machine Learning, Interworks